

Intrapartum Sildenafil to Improve Perinatal Outcomes A Randomized Clinical Trial

Sailesh Kumar, DPhil(Oxon); William Tarnow-Mordi, MD; Ben W. Mol, PhD; Vicki Flenady, PhD; Helen G. Liley, MD; Nadia Badawi, PhD; Susan Walker, MD; Jonathan Hyett, MD; Anna Lene Seidler, PhD; Emily Callander, PhD; John Simes, MD; Rachel L. O'Connell, PhD; for the iSEARCH Investigators

IMPORTANCE Sildenafil citrate may increase uteroplacental blood flow. Its ability to reduce perinatal complications related to fetal hypoxia during labor is uncertain.

OBJECTIVE To compare the effectiveness of intrapartum maternal oral sildenafil citrate vs placebo in improving perinatal outcomes potentially related to intrapartum hypoxia in term pregnancies.

DESIGN, SETTING, AND PARTICIPANTS This pragmatic, multicenter, investigator-initiated, placebo-controlled randomized clinical trial including 3257 women was conducted in 13 Australian hospitals from September 6, 2021, to June 28, 2024. The last date of follow-up (28-day neonatal mortality) was July 26, 2024. Women aged 18 years or older with singleton or dichorionic twin pregnancies, planning vaginal birth at term by either spontaneous labor or induction of labor, were recruited.

INTERVENTIONS Women were assigned to 50 mg oral sildenafil citrate every 8 hours up to 150 mg or equivalent placebo.

MAIN OUTCOME AND MEASURES The primary composite outcome was intrapartum stillbirth, neonatal death, Apgar score less than 4 at 5 minutes (a score of <4 at 5 minutes is indicative of severe neonatal depression at birth, with scores ranging from 0 to 10), acidosis at birth (umbilical cord artery pH <7.0), hypoxic ischemic encephalopathy, neonatal seizures, neonatal respiratory support for greater than 4 hours, neonatal unit admission for greater than 48 hours, persistent pulmonary hypertension of the newborn, or meconium aspiration syndrome. Secondary outcomes were the individual components of the primary composite and emergency cesarean delivery or instrumental birth for intrapartum fetal distress.

RESULTS A total of 3257 women were randomized to sildenafil citrate (n = 1626 women and 1634 infants) or placebo (n = 1631 women and 1641 infants). Mean (SD) maternal age and gestation at randomization were similar in both groups (31.7 [5.1] vs 31.5 [5.0] years and 39.5 [1.2] vs 39.5 [1.1] weeks, respectively). A total of 868 participants (53.4%) vs 874 participants (53.6%) were of Australia/New Zealand ethnicity and 315 participants (19.4%) vs 311 participants (19.1%) were of European ethnicity. Most participants were nulliparous (944 of 1624 [58.1%; 2 missing values] vs 966 of 1630 [59.3%; 1 missing value]). Induction of labor occurred in 1353 of 1621 women (83.5%) in the sildenafil citrate group and 1348 of 1627 women (82.9%) in the placebo group. The primary outcome occurred in 83 of 1625 women (5.1%) in the sildenafil citrate group and 84 of 1625 (5.2%) in the placebo group (relative risk, 1.02; 95% CI, 0.75-1.37). Sildenafil citrate had no significant effect on emergency cesarean delivery or instrumental vaginal birth for fetal distress (relative risk, 1.12; 95% CI, 0.98-1.29) or on any of the individual components of the primary outcome. Subgroup analyses showed no evidence of heterogeneity of treatment effect.

CONCLUSIONS AND RELEVANCE Sildenafil citrate did not result in a lower incidence of adverse perinatal outcomes potentially related to intrapartum hypoxia.

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Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The iSEARCH investigators appear in Supplement 4.

Corresponding Author: Sailesh Kumar, DPhil(Oxon), Mater Research Institute–University of Queensland, Mater Campus, Whitty Building, Level 1, South Brisbane, Queensland 4101, Australia (sailesh.kumar@mater.uq.edu.au).

During labor, uterine contractions cause intermittent but progressive reduction in placental perfusion, which gradually limits fetal oxygenation. Intrapartum fetal compromise (fetal distress) occurs when there is inadequate placental reperfusion time between contractions or when there is appropriate reperfusion time but underlying placental dysfunction limits adequate oxygen transfer to the fetus.¹ Hypoxic-ischemic injury during labor^{2,3} is an important cause of intrapartum stillbirth, neonatal death, and neonatal encephalopathy in late preterm and term infants, with small for gestational age infants at particular risk.⁴ Although the prevalence of hypoxia-related birth injury is high in poorly resourced settings, the proportion of true intrapartum hypoxic-ischemic events causing neonatal encephalopathy is low in many high-income countries.⁵ When fetal distress is suspected, emergency cesarean delivery or instrumental vaginal birth is often required to prevent ongoing in utero injury to the infant. Although these urgent interventions may be necessary, they can be associated with significant maternal and neonatal morbidity and have implications for future pregnancies.⁶

Criteria used by various international bodies to determine whether an intrapartum hypoxic event is responsible for neonatal encephalopathy include sentinel events occurring during labor, specific pathologic fetal heart rate patterns, neuroimaging abnormalities consistent with an acute intrapartum event, subsequent spastic quadriplegia or dyskinetic cerebral palsy, and exclusion of other causative or contributory factors.^{7,8} However, although electronic fetal heart rate monitoring during labor is often used to detect fetal compromise, its use has not resulted in improvements in cerebral palsy rates, perinatal mortality, or other measures of neonatal well-being,⁹ emphasizing the imperative for more effective or novel approaches to prevent adverse perinatal outcomes related to intrapartum hypoxic injury. One possible strategy is to improve fetoplacental perfusion and fetal oxygenation during labor.

Cyclic nucleotide phosphodiesterases (PDEs) are a family of 11 related phosphohydrolase isoforms that selectively catalyze the hydrolysis of the 3' cyclic phosphate bonds of adenosine or guanosine 3',5' cyclic monophosphate,¹⁰ increasing bioavailability of nitric oxide causing vasodilatation. Phosphodiesterase type 5 (PDE5) inhibitors are licensed for the treatment of erectile dysfunction in men¹¹ and severe pulmonary arterial hypertension in adults, neonates, and children.^{12,13} High levels of PDE5 are found in vascular smooth muscle and placenta. In human and animal models, both in vivo and in vitro data suggest that PDE5 inhibitors increase uteroplacental perfusion¹⁴⁻¹⁶ and reverse vasoconstriction in uterine and spiral arteries.^{17,18} Only sildenafil citrate and tadalafil have been used for a variety of indications related to placental dysfunction, to treat maternal hypertension, or both.¹⁹

In an earlier phase 2 randomized clinical trial (RCT),²⁰ we showed that maternal oral sildenafil citrate compared with placebo reduced the relative risk (RR) of operative birth for fetal distress by 51% (18% vs 36.7%; RR, 0.49; 95% CI, 0.33-0.73). It also attenuated the normal intrapartum decline in placental growth factor levels, suggesting that it may preserve pla-

Key Points

Question Does maternal oral sildenafil citrate during labor improve perinatal outcomes potentially related to intrapartum hypoxia?

Findings This placebo-controlled randomized trial of 3257 women found no difference in the frequency of the primary composite outcome of infant mortality or neonatal morbidity between women assigned to receive oral sildenafil citrate in labor compared with placebo (5.1% vs 5.2%).

Meaning Sildenafil citrate did not result in a lower incidence of adverse perinatal outcomes after term labor and should not be used for this indication.

cental function in labor.²¹ This trial, however, lacked power to show an effect on perinatal outcomes related to intrapartum hypoxic injury.²⁰

We report the results of the Australian iSEARCH (Can Intrapartum Sildenafil Safely Avert the Risks of Contraction-Induced Hypoxia in Labor) RCT, which tested the hypothesis that, compared with placebo, intrapartum maternal oral sildenafil citrate improves perinatal outcomes in term pregnancies.

Methods

Study Design and Oversight

iSEARCH was a pragmatic, multicenter, investigator-initiated, placebo-controlled, double-blind RCT involving women with term pregnancies (≥ 37 weeks' gestation) attempting vaginal birth at 13 Australian hospitals from September 6, 2021, to June 28, 2024. The trial protocol has been published.²² The trial was approved by the Hunter New England Local Health District Human Research Ethics Committee, New South Wales Ministry of Health.

Participants

Women aged 18 years or older with singleton or dichorionic twin pregnancies, planning vaginal birth at term by either spontaneous labor or induction of labor, were eligible for the trial. Women with triplets or higher-order multifetal gestation, with monochorionic twins, taking any type of nitrate-containing medications or other PDE inhibitors, or affected by severe hepatic or kidney impairment were excluded.

Screening, Recruitment, and Randomization

Women at at least 34 weeks' gestation at all participating hospitals were screened for eligibility by study midwives, and those who met the inclusion criteria and consented to participate were registered in an online trial registration database (Medidata Solutions, Dassault Systèmes). Once registered, each woman was assigned a unique study number and continued to receive routine prenatal care until spontaneous labor or induction of labor occurred. Written informed consent was obtained from all women at registration before randomization.

After spontaneous labor had been diagnosed or after artificial rupture of membranes if induction of labor was

undertaken, randomization was performed by the responsible midwife or obstetrician after confirming that the woman met the eligibility criteria. Only women who were in spontaneous labor or undergoing induction at term (≥ 37 weeks) were randomized. Women who were registered but who delivered before 37 weeks were not randomized and were not part of the analysis of study outcomes. Randomization was performed using a web-based (Medidata Solutions), computer-generated, concealed allocation sequence, stratified by study site. After randomization, women were assigned a treatment group. The first participant was registered on September 6, 2021; and the final participant, on June 28, 2024.

Trial Interventions and Management of Labor and Puerperium

Women were assigned in a 1:1 ratio to receive 50 mg sildenafil citrate or identical placebo orally every 8 hours to a maximum of 3 doses. Trial medication was manufactured by Optima Ovest. The Therapeutic Goods Administration is Australia's government authority responsible for evaluating, assessing, and monitoring products that are defined as therapeutic goods for use in clinical practice or research studies. Trial medication was stored at room temperature at each center's pharmacy and transferred to the labor ward when required.

All women, regardless of treatment group, were treated according to local intrapartum and postnatal protocols and guidelines (including the use of electronic fetal heart rate monitoring).^{23,24} Spontaneous labor was defined as regular uterine contractions with progressive cervical dilatation.²³ Methods for induction of labor included vaginal prostaglandins, balloon cervical dilatation, artificial rupture of membranes, and oxytocin infusion.²³

Classification of fetal heart rate abnormalities followed Royal Australian and New Zealand College of Obstetricians and Gynecologists guidelines.²⁵ Where possible, umbilical artery cord pH was measured after birth. To detect persistent pulmonary hypertension of the newborn, all neonates received oxygen saturation screening 24 to 48 hours after birth.

Trial Outcomes

The primary outcome was a composite of 10 intrapartum or neonatal events potentially reflecting intrapartum hypoxia, including 1 or more of intrapartum stillbirth, 28-day neonatal death, Apgar score less than 4 at 5 minutes (a score of <4 at 5 minutes is indicative of severe neonatal depression at birth, with scores ranging from 0 to 10), acidosis at birth (umbilical cord artery pH <7.0), hypoxic ischemic encephalopathy, neonatal seizures, neonatal respiratory support for greater than 4 hours, neonatal unit admission for greater than 48 hours, persistent pulmonary hypertension of the newborn, or meconium aspiration syndrome. Similar (but not identical) composite outcomes have been used in other large perinatal trials^{26,27} to reflect severe neonatal outcome (death or severe morbidity). Secondary outcomes included each of the 10 individual components of the primary composite outcome and emergency cesarean delivery or instrumental vaginal birth for fetal distress. Tertiary outcomes were emergency cesarean

delivery for fetal distress (primary indication), instrumental vaginal birth for fetal distress (primary indication), cesarean delivery for indications other than fetal distress, instrumental vaginal birth for indications other than fetal distress, spontaneous vaginal birth, length of first and second stage of labor, postpartum hemorrhage greater than 1 L, postpartum hysterectomy, uterine rupture, maternal intensive care unit admission, and maternal death. Primary and secondary outcomes were prespecified in the trial protocol (Supplement 1) and prospective trial registration records; additional hypothesis-generation tertiary outcomes were prespecified in the statistical analysis plan (Supplement 2).

Cases that met the primary and secondary outcomes were reviewed by the local lead investigator. In-hospital maternal and neonatal outcome events were collected from medical records into electronic case report forms. Apart from 28-day neonatal mortality, all primary outcome data were routinely available before discharge and collected electronically. There were no formal study assessments visits.

An independent data and safety monitoring committee conducted 1 interim analysis after 50% of participants were recruited. A recommendation to stop the trial early required a difference in perinatal mortality indicated by a Haybittle-Peto boundary of 3 SDs from the null (equivalent to $P < .003$).^{28,29}

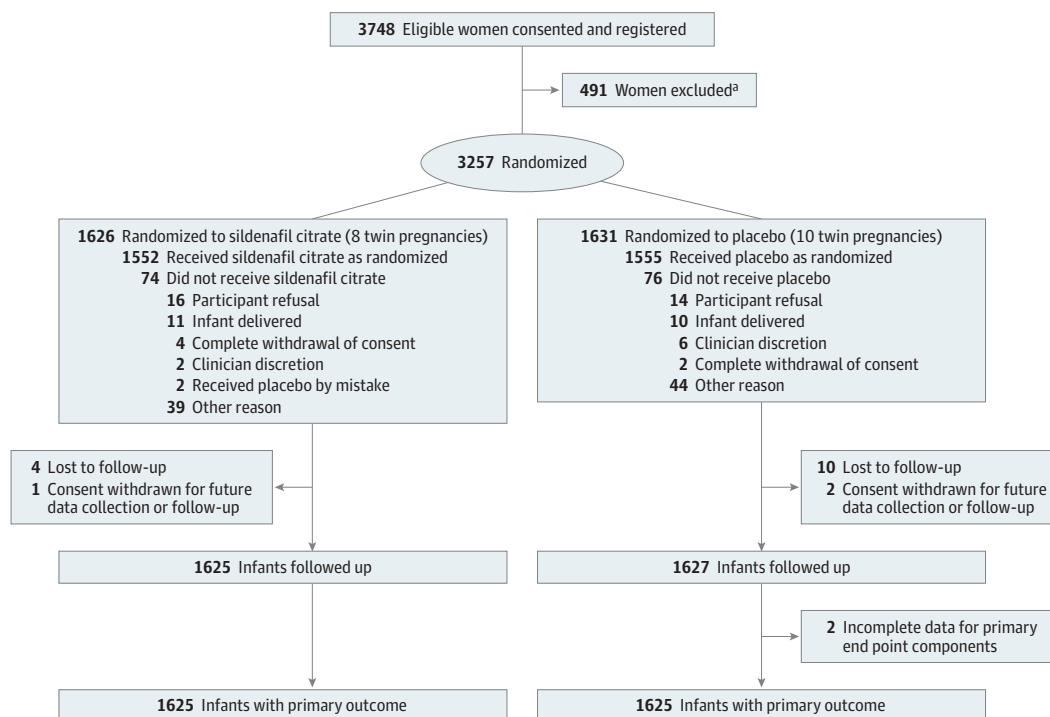
Sample Size Calculation

The incidence of the primary composite adverse outcome was estimated as 7% based on data from participating hospitals and the ARRIVE trial.²⁶ A sample size of 3200 women (allowing for 10% lost to follow-up) permitted detection of a 35% reduction in the primary outcome (from 7.0% to 4.6%) for an α of .05 and greater than 80% power in the treatment group compared with the placebo group. This sample size also yielded greater than 90% power to detect a 25% reduction for the secondary outcome of cesarean delivery or instrumental vaginal birth for fetal distress.

Statistical Analysis

Analyses were performed according to the intention-to-treat principle and were prespecified in the trial protocol (Supplement 1) and statistical analysis plan (Supplement 2) before the trial database was locked and unblinding performed. The primary composite outcome was analyzed using a generalized estimating equation log-binomial model adjusting for clustering due to multifetal gestation, with RR and 95% CI to indicate the relative reduction (or increase) in risk associated with sildenafil citrate therapy. χ^2 Tests (with linear trend test for ordered variables) were used to compare categorical variables and t tests were used for continuous variables, or if the distribution of the data was nonnormal, the Wilcoxon rank sum test was used. Secondary analysis of the primary outcome was undertaken, which included adjustment for prognostic factors using robust (modified) Poisson regression. Because the secondary outcomes are interrelated and potentially causally attributable to hypoxic birth injury, we interpreted the pattern and consistency of findings across related outcomes rather than focusing on any single, statistically significant result.³⁰ Thus, results from secondary outcomes and

Figure 1. Randomization, Delivery, and Assessment of Patients in a Clinical Trial of Sildenafil



^aReasons women were excluded are not shown because this information was not routinely recorded by participating sites.

subgroup analyses should not be used to infer definitive treatment effect.

Prespecified subgroup analyses were performed by assessing interaction effects on the primary composite outcome. We prespecified subgroup analyses for women with small for gestational age infants, multifetal gestation, preeclampsia, noncephalic presentation, cesarean delivery in preceding pregnancy, and postnatal diagnosis of fetal anomaly. Because cord artery pH testing was performed for only 1223 of 1629 women (75.1%) in the sildenafil citrate group and 1159 of 1637 women (70.8%) in the placebo cohort (5 and 4 missing values, respectively, because of withdrawals and loss to follow-up), the primary outcome was evaluated with this component imputed several ways. In the primary analysis, no event was assumed when cord artery pH testing was not done. Because 1 center (Mater Mother's Hospital) recruited 54.1% of the trial cohort (1763 of 3257), we undertook additional post hoc analyses for the primary, secondary, and tertiary outcomes between the Mater Mother's Hospital and other participating sites. Safety analysis of adverse events was undertaken among patients who received at least 1 dose of study treatment, with subjects counted in the group according to the treatment they received. Two-tailed $P < .05$ was considered statistically significant. There was no adjustment for multiple testing. Statistical analyses were performed with SAS version 9.4 (SAS Institute); data were analyzed on October 24, 2024. Reporting followed CONSORT³¹ and TRIPOD³² guidelines.

Results

Participants and Adherence

Between September 6, 2021, and June 28, 2024, 3257 women were randomized after written informed consent (1626 receiving sildenafil citrate [1634 infants] and 1631 allocated to placebo [1641 infants]). After randomization 9 women withdrew consent, leaving 3248 women (3266 infants) with outcome data for the final intention-to-treat analysis. The primary end point was available for 3250 infants (Figure 1). The last date of follow-up (28-day neonatal mortality) was July 26, 2024.

Baseline characteristics for the 2 groups were comparable. Mean maternal age and gestation at randomization were 31.7 years (SD, 5.1 years) and 39.5 weeks (SD, 1.2 weeks), respectively, in the sildenafil citrate group and 31.5 years (SD, 5.0 years) and 39.5 weeks (SD, 1.1 weeks) in the placebo cohort. Ethnicity of the 1626 participants in the sildenafil citrate group and 1631 participants in the placebo cohort included Aboriginal and Torres Strait Islander (47 [2.9%] vs 42 [2.6%]), African (14 [0.9%] vs 18 [1.1%]), Australia/New Zealand (868 [53.4%] vs 874 [53.6%]), European (315 [19.4%] vs 311 [19.1%]), missing (2 [0.1%] vs 2 [0.1%]), other (68 [4.2%] vs 73 [4.5%]; no further breakdown was available for this category), Pacific Islander (53 [3.3%] vs 53 [3.2%]), and Southeast Asian/Northeast Asian/South Asian (241 [14.8%] vs 244 [15.0%]). Ethnicity was a self-reported variable in

Table 1. Characteristics of Study Participants^a

	Sildenafil citrate (n = 1626)	Placebo (n = 1631)
Age, mean (SD), y	31.7 (5.1)	31.5 (5.0)
Gestation at randomization, mean (SD), wk	39.5 (1.2)	39.5 (1.1)
Ethnicity ^b	n = 1624	n = 1629
Aboriginal and Torres Strait Islander	47 (2.9)	42 (2.6)
African	14 (0.9)	18 (1.1)
Australia/New Zealand	868 (53.4)	874 (53.7)
European	315 (19.4)	311 (19.1)
Middle Eastern	18 (1.1)	14 (0.9)
Other ^c	68 (4.2)	73 (4.5)
Pacific Islander	53 (3.3)	53 (3.3)
Southeast Asian/Northeast Asian/South Asian	241 (14.8)	244 (15.0)
BMI at first hospital visit, median (IQR)	25.9 (22.4-31.2)	25.7 (22.4-30.5)
Smoking during pregnancy	77 (4.7) [n = 1624]	72 (4.4) [n = 1630]
Parity (births >20 wk gestation)	n = 1624	n = 1630
0	944 (58.1)	966 (59.3)
1	446 (27.5)	450 (27.6)
2	152 (9.4)	153 (9.4)
≥3	82 (5.0)	61 (3.7)
Twin pregnancy	8 (0.5) [n = 1621]	10 (0.6) [n = 1627]
Method of birth in preceding pregnancy	n = 680	n = 664
Spontaneous vaginal birth	477 (70.1)	432 (65.1)
Instrumental vaginal birth	146 (21.5)	159 (23.9)
Emergency cesarean delivery	49 (7.2)	51 (7.7)
Elective cesarean delivery	8 (1.2)	22 (3.3)
Mode of conception	n = 1624	n = 1630
Natural	1461 (90.0)	1468 (90.1)
Assisted	163 (10.0)	162 (9.9)
Hypertension	n = 1624	n = 1630
Gestational	84 (5.2)	83 (5.1)
Preeclampsia	41 (2.5)	39 (2.4)
Chronic/essential	32 (2.0)	24 (1.5)
Diabetes mellitus	n = 1624	n = 1630
Gestational	318 (19.6)	324 (19.9)
Type I and II	29 (1.8)	25 (1.5)
Antepartum hemorrhage	11 (0.7) [n = 1624]	7 (0.4) [n = 1630]
Small for gestational age	56 (3.4) [n = 1624]	56 (3.4) [n = 1630]
No. of trial medication doses received		
1	902 (55.5)	914 (56.0)
2	558 (34.3)	549 (33.7)
3	94 (5.8)	92 (5.6)
Recruitment at participating hospitals		
Mater Mother's Hospital	880 (54.1)	883 (54.1)
Other	476 (29.3)	476 (29.2)
Royal Prince Alfred Hospital	164 (10.1)	166 (10.2)
Royal Brisbane and Women's Hospital	106 (6.5)	106 (6.5)

Abbreviation: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

^a Data are presented as No. (%) unless otherwise indicated.

^b Ethnicity was a self-reported variable in response to an open-ended question.

^c No further breakdown was available for this category.

response to an open-ended question. In the sildenafil citrate cohort, 902 (55.5%), 558 (34.3%), and 94 (5.8%) women received 1, 2, and 3 doses, respectively; in the placebo group, 914 (56.0%), 549 (33.7%), and 92 (5.6%) women received 1, 2, and 3 doses, respectively (Table 1). Most participants were nulliparous (944 of 1624 [58.1%; 2 missing values] vs 966 of 1630 [59.3%; 1 missing value]).

Mean gestation at birth was similar in both groups (sildenafil citrate, 39.6 weeks [SD, 1.2]; placebo, 39.6 weeks [SD, 1.2]). Induction of labor occurred for 83.5% of women (1353 of 1621) in the sildenafil citrate group and 82.9% of women (1348 of 1627) receiving placebo. The proportion of women who had electronic fetal heart rate monitoring in labor was similar in both cohorts (1579 [97.4%] vs 1585 [97.4%]) (Table 2).

Table 2. Intrapartum and Birth Outcomes^a

	Sildenafil citrate	Placebo
Neonates, No.	1634	1641
Gestation at birth, mean (SD), wk	39.6 (1.2)	39.6 (1.2)
Infant sex	n = 1629	n = 1637
Male	833 (51.1)	840 (51.3)
Female	796 (48.9)	797 (48.7)
Birth weight, mean (SD), g	3451.9 (474.9)	3466.6 (452.7)
Mode of birth	n = 1629	n = 1637
Spontaneous vaginal birth	864 (53.0)	931 (56.9)
Cesarean delivery	466 (28.6)	412 (25.2)
Instrumental vaginal birth (vacuum)	209 (12.8)	204 (12.5)
Instrumental vaginal birth (forceps)	86 (5.3)	88 (5.4)
Vaginal breech birth	4 (0.2)	2 (0.1)
Women, No.	1626	1631
Onset of labor	n = 1621	n = 1627
Induction	1353 (83.5)	1348 (82.9)
Spontaneous	268 (16.5)	279 (17.1)
Method of induction of labor		
Oxytocin with or without artificial rupture of membranes	1252 (92.5)	1227 (91.0)
Balloon cervical dilatation	562 (41.5)	592 (43.9)
Prostaglandin	465 (34.4)	474 (35.2)
Artificial rupture of membranes only	60 (4.4)	76 (5.6)
Details of labor		
Meconium-stained liquor	294 (18.1) [n = 1621]	268 (16.5) [n = 1626]
Analgesia in labor	n = 1621	n = 1625
Epidural/spinal	1239 (76.4)	1241 (76.4)
Nitrous oxide (gas and air)	217 (13.4)	221 (13.6)
None	133 (8.2)	127 (7.8)
IM pethidine or morphine	32 (2.0)	36 (2.2)
Electronic fetal monitoring in labor	1579 (97.4) [n = 1621]	1585 (97.4) [n = 1627]
Length of stage 1, median (range), h	3.58 (0.00-30.58)	3.55 (0.00-40.08)
Length of stage 2, median (range), h	0.40 (0.00-49.00)	0.43 (0.00-33.00)
Complications during labor		
Suspected abruption	5 (0.3) [n = 1621]	9 (0.6) [n = 1627]
Intrapartum bleeding	17 (1.0) [n = 1621]	17 (1.0) [n = 1627]
Cord prolapse	2 (0.1) [n = 1621]	3 (0.2) [n = 1627]
Uterine rupture	4 (0.2) [n = 1621]	2 (0.1) [n = 1627]
Clinical chorioamnionitis	28 (1.7) [n = 1621]	20 (1.2) [n = 1627]
Maternal eclampsia/seizures	0	0
Postpartum hemorrhage (>1 L)	164 (10.1) [n = 1621]	128 (7.9) [n = 1627]
Postpartum hysterectomy	1 (0.1) [n = 1621]	1 (0.1) [n = 1627]
Maternal intensive care unit admission	2 (0.1) [n = 1617]	1 (0.1) [n = 1614]
Length of hospital stay, median (range), d	3 (0-62)	3 (0-41)

Abbreviation: IM, intramuscular.

^a Data are presented as No. (%) unless otherwise indicated.

Primary and Secondary Outcomes

The primary composite outcome occurred in 83 of 1625 women (5.1%) in the sildenafil citrate group and 84 of 1625 women (5.2%) in the placebo group (RR, 1.02; 95% CI, 0.75-1.37 after adjustment for clustering due to multifetal gestation) (Table 3). There were no cases of infant death. Umbilical cord artery pH testing was performed for only 1223 of 1629 women (75.1%) in the sildenafil citrate group and 1159 of 1637 women (70.8%) in the placebo cohort (5 and 4 missing values, respectively, owing to withdrawals or loss to

follow-up). However, the finding for the primary outcome did not change materially whether missing cord artery pH was either imputed (RR, 1.01; 95% CI, 0.75-1.36) or adjusted for (RR, 0.98; 95% CI, 0.73-1.31) (eTable 1 in Supplement 3). Sildenafil citrate had no effect on emergency operative birth (cesarean delivery or instrumental vaginal birth) for fetal distress (RR, 1.12; 95% CI, 0.98-1.29) nor on any of the individual components of the primary outcome: Apgar score less than 4 at 5 minutes (RR, 1.67; 95% CI, 0.40-7.00), cord artery pH less than 7 (RR, 2.28; 95% CI, 0.81-6.47), neonatal

Table 3. Primary and Secondary Outcomes

	No./total No. (%)		Sildenafil citrate vs placebo (95% CI)		P value ^a
	Sildenafil citrate	Placebo	Absolute difference, %	Relative risk ^a	
Primary outcome	83/1625 (5.1)	84/1625 (5.2)	-0.1 (-1.6 to 1.5)	1.02 (0.75 to 1.37)	.91
Secondary outcomes					
Intrapartum stillbirth	0/1629	0/1637	NA	NA	NA
28-d Neonatal death	0/1625	0/1627	NA	NA	NA
Apgar score <4 at 5 min	5/1629 (0.3)	3/1637 (0.2)	0.1 (-0.3 to 0.5)	1.67 (0.40 to 7.00)	.48
Umbilical cord artery pH <7.0 ^b	12/1629 (0.7)	5/1637 (0.3)	0.4 (-0.1 to 1.0)	2.28 (0.81 to 6.47)	.12
Neonatal encephalopathy	1/1625 (0.1)	1/1625 (0.1)	0.0 (-0.2 to 0.2)	0.94 (0.06 to 14.94)	.96
Neonatal seizures within 28 d	2/1625 (0.1)	1/1625 (0.1)	0.1 (-0.2 to 0.3)	1.87 (0.17 to 20.62)	.61
Neonatal unit admission >48 h	50/1629 (3.1)	61/1637 (3.7)	-0.7 (-2.0 to 0.6)	0.85 (0.59 to 1.23)	.39
Neonatal respiratory support >4 h	37/1629 (2.3)	47/1637 (2.9)	-0.6 (-1.7 to 0.5)	0.81 (0.53 to 1.25)	.35
Persistent pulmonary hypertension of the newborn	1/1629 (0.1)	3/1637 (0.2)	-0.1 (-0.4 to 0.2)	0.33 (0.03 to 3.22)	.34
Meconium aspiration	9/1629 (0.6)	5/1637 (0.3)	0.2 (-0.3 to 0.8)	1.81 (0.61 to 5.39)	.29
Emergency cesarean delivery or instrumental vaginal birth for fetal distress	343/1629 (21.1)	307/1637 (18.8)	2.3 (-0.5 to 5.1)	1.12 (0.98 to 1.29)	.10

Abbreviation: NA, not applicable.

^a Generalized estimating equations adjusting for multiple births.

^b Not done for cord artery pH less than 7 assumed to be no event.

encephalopathy (RR, 0.94; 95% CI, 0.06-14.94), neonatal seizures (RR, 1.87; 95% CI, 0.17-20.62), neonatal unit admission (RR, 0.85; 95% CI, 0.59-1.23), neonatal respiratory support greater than 4 hours (RR, 0.81; 95% CI, 0.53-1.25), persistent pulmonary hypertension of the newborn (RR, 0.33; 95% CI, 0.03-3.22), or meconium aspiration syndrome (RR, 1.81; 95% CI, 0.61-5.39) (Table 3).

Subgroup analyses related to small for gestational age infants, multifetal gestation, preeclampsia, noncephalic presentation, preceding pregnancy cesarean delivery, and postnatal diagnosis of fetal anomaly showed no appreciable differences in treatment effect across the subgroup categories, consistent with the overall result of no effect (Figure 2).

Other Maternal and Neonatal Outcomes

There were no differences in length of hospital stay, maternal intensive care unit admission, length of labor, method of analgesia, or use of electronic fetal heart rate monitoring. Rates of meconium-stained liquor, abruption, intrapartum bleeding, cord prolapse, uterine rupture, chorioamnionitis, or hysterectomy were also no different between the sildenafil citrate and placebo groups (Table 4).

There were no significant differences between the 2 groups regarding prespecified adverse effects potentially related to sildenafil or any other serious maternal or neonatal outcomes. Specifically, there were no cases of maternal mortality, intrapartum stillbirth, or neonatal death in either group. One case of persistent pulmonary hypertension occurred in the sildenafil citrate group compared with 3 cases in the placebo group (eTable 3 in Supplement 3).

Site-Specific Differences

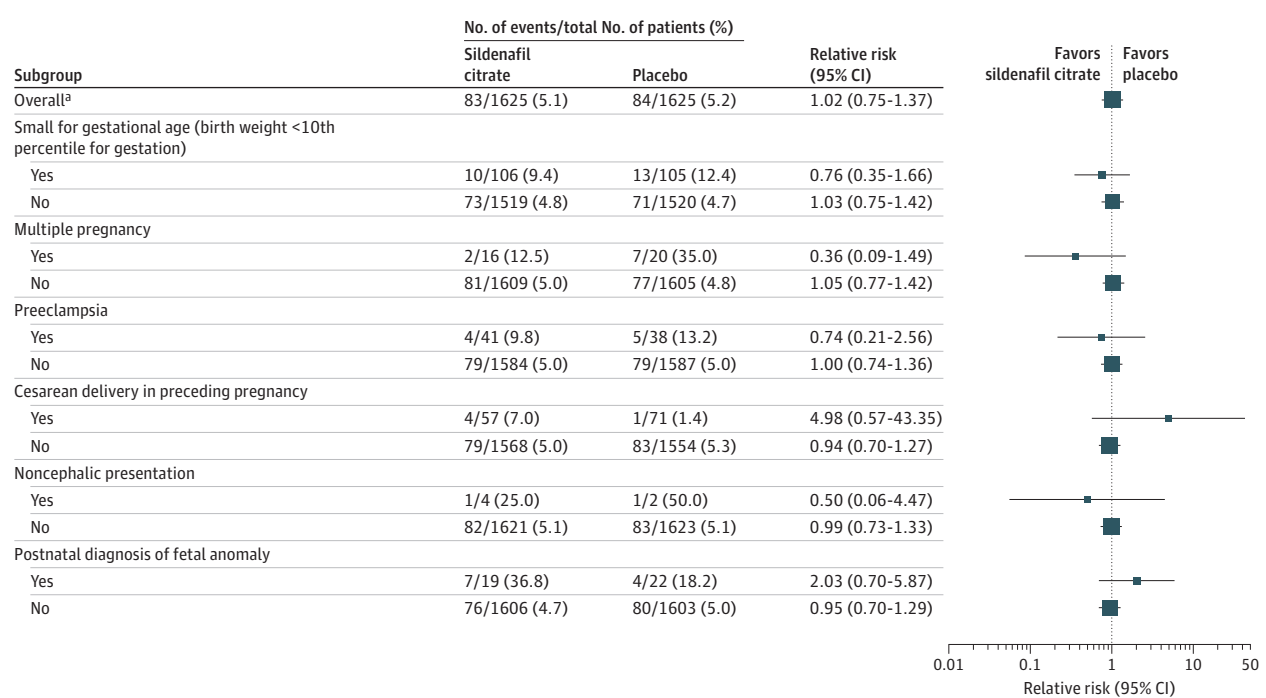
There were potentially important site-specific differences in outcomes: the primary outcome at the Mater Mother's Hospi-

tal occurred in 16 of 883 women (1.8%) in the sildenafil citrate group and 24 of 887 women (2.7%) in the placebo group compared with 67 of 742 women (9.0%) and 60 of 738 women (8.1%), respectively, at the other sites. Rates of emergency operative birth (cesarean delivery or instrumental vaginal birth) for fetal distress were also lower at the Mater Mother's Hospital compared with that at the other sites (eTable 2 in Supplement 3). Although rates of postpartum hemorrhage, which was 1 of 11 hypothesis-generating tertiary outcomes, were higher in the sildenafil citrate group (164 of 1621 [10.1%] vs 128 of 1627 [7.9%]; absolute risk difference, 2.2%; 95% CI, 0.2%-4.3%), there was no difference in this outcome at the Mater Mother's Hospital between the 2 groups (73 of 880 [8.3%] vs 72 of 882 [8.2%]) (eTable 2 in Supplement 3).

Discussion

In this randomized trial of women attempting vaginal birth at term, we did not find any difference in the frequency of the primary composite outcome of infant mortality or neonatal morbidity between those assigned to receive oral sildenafil citrate in labor compared with placebo. There were also no differences in the secondary outcomes of emergency operative birth for fetal distress nor for any of the components of the primary composite outcome. There were important nonsignificant differences in frequency of outcomes between the center with the largest number of participating women (Mater Mother's Hospital, 1763 of 3257 [54.1%]) and other hospitals. Post hoc analyses comparing outcomes between the Mater Mother's Hospital and other centers showed that the frequency of the primary composite outcome in the sildenafil citrate group was 33% lower compared with placebo (16 of 883 [1.8%] vs 24 of 887

Figure 2. Prespecified Subgroup Analyses According to Maternal and Neonatal Characteristics



The forest plot shows relative risks and 95% CIs in a comparison of sildenafil citrate with placebo on the composite primary end point. The size of the squares is proportional to the amount of statistical information in that category (ie, the inverse of the variance). Interaction tests between treatment and subgroup were not significant ($P > .05$).

^aThe overall treatment effect is based on a generalized estimating equation log-binomial model. Effects within subgroups were estimated using a generalized linear log-binomial model because the analysis was exploratory and because of the small number of twins within subgroups.

Table 4. Tertiary Outcomes

	No./total No. (%)		Absolute difference (95% CI), %	Relative risk/difference in means (95% CI) ^a
	Sildenafil citrate	Placebo		
Infant-level analyses, No.	1634	1641	NA	NA
Cesarean delivery for fetal distress	142/1629 (8.7)	129/1637 (7.9)	0.8 (−1.1 to 2.8)	1.11 (0.88 to 1.39)
IVB for fetal distress	201/1629 (12.3)	178/1637 (10.9)	1.5 (−0.8 to 3.7)	1.13 (0.94 to 1.37)
Cesarean delivery for other indications	291/1629 (17.9)	257/1637 (15.7)	2.2 (−0.5 to 4.8)	1.14 (0.97 to 1.32)
IVB for other indications	94/1629 (5.8)	114/1637 (7.0)	−1.2 (−2.9 to 0.5)	0.83 (0.64 to 1.08)
Emergency operative birth (fetal distress was a factor)	471/1629 (28.9)	431/1637 (26.3)	2.6 (−0.5 to 5.7)	1.10 (0.98 to 1.23)
Cesarean delivery (fetal distress was a factor)	233/1629 (14.3)	201/1637 (12.3)	2.0 (−0.4 to 4.4)	1.16 (0.98 to 1.39)
IVB (fetal distress was a factor)	238/1629 (14.6)	230/1637 (14.1)	0.6 (−1.9 to 3.0)	1.04 (0.88 to 1.23)
Spontaneous vaginal birth	864/1629 (53.0)	931/1637 (56.9)	−3.8 (−7.3 to −0.4)	0.93 (0.88 to 0.99)
Maternal-level analyses, No.	1626	1631	NA	NA
Length of stage 1 labor, mean (SD), h	4.53 (4.19)	4.53 (4.14)	NA	−0.00 (−0.29 to 0.28)
Length of stage 2 labor, mean (SD), h	0.85 (1.59)	0.88 (1.39)	NA	−0.03 (−0.14 to 0.07)
Postpartum hemorrhage >1 L	164/1621 (10.1)	128/1627 (7.9)	2.2 (0.2 to 4.3)	1.29 (1.03 to 1.60)
Postpartum hysterectomy	1/1621 (0.1)	1/1627 (0.1)	0.0 (−0.2 to 0.2)	1.00 (0.06 to 16.03)
Uterine rupture	4/1621 (0.2)	2/1627 (0.1)	0.1 (−0.2 to 0.5)	2.01 (0.37 to 10.94)
Maternal intensive care unit admission	2/1617 (0.1)	1/1614 (0.1)	0.1 (−0.2 to 0.3)	2.00 (0.18 to 21.99)
Maternal mortality	0/1617	0/1618	NA	NA

Abbreviations: IVB, instrumental vaginal birth; NA, not applicable.

^a For infant-level analyses, relative risk estimates were calculated with generalized estimating equations accounting for multiple births.

[2.7%]) vs 11% higher (67 of 742 [9.0%] vs 60 of 738 [8.1%]) at the other sites, but this difference across sites was still nonsignificant (interaction $P = .16$).

The findings of this RCT are thus inconsistent with the promising results of an earlier single-center phase 2 trial²⁰ that showed a 51% reduction in emergency operative birth rates for intrapartum fetal distress and the findings of a subsequent systematic review and meta-analysis¹⁹ of 1090 women that found that PDE5 inhibitor use was associated with a 42% relative reduction of the same outcome. One explanation may be that in the previous phase 2 study, the investigators independently reviewed fetal heart rate recordings and indications for emergency operative birth for all trial participants and recategorized these ratings, if appropriate. However, these steps were not taken in the current RCT (given its pragmatic nature), and indications for operative birth as judged by the treating obstetric team were accepted without further review. It is therefore possible that in some cases in which an element of significant fetal compromise was present, it was not recorded as the primary reason for emergency operative birth because of another competing indication. Alternatively, the results in the previous trial may have been a chance finding, demonstrating the importance of large phase 3 trials to confirm findings of early promising studies.

Phosphodiesterase type 5 inhibitors may prevent intrapartum fetal distress by increasing fetoplacental blood flow,¹⁴⁻¹⁶ reversing vasoconstriction in uterine and spiral arteries,^{17,18} and preserving intrapartum placental function and fetoplacental perfusion,²¹ thereby increasing fetal oxygenation and tolerance to progressive intrapartum hypoxia. However, many of the data for PDE5 inhibitor use in pregnancy relate to treatment of severe preterm fetal growth restriction.^{19,33} Because sildenafil citrate for this indication is not efficacious and may increase the risk of persistent pulmonary hypertension of the newborn, it is not recommended for the treatment of fetal growth restriction and should not be used outside of a research setting.³⁴⁻³⁶ Our hypothesis-generating finding of a possible increase in the risk of postpartum hemorrhage may be a type I error related to multiple comparisons but emphasizes the need for caution: through its relaxant action on myometrial smooth muscle,³⁷ uterine atony may be aggravated, increasing the risk of bleeding.

Strengths of the study include its size, high level of adherence to the protocol, and low levels of missing data. The primary perinatal outcome included mortality and other outcomes identified by parents, former patients, health care professionals, and researchers as important.³⁸ This RCT also demonstrates the feasibility of intrapartum randomization of women and that it is possible to recruit for and conduct a large pharmacologic study even in low-risk pregnancies, notwithstanding the obvious challenges of therapeutic trials in pregnancy.^{39,40}

Limitations

This RCT has several limitations. First, the overall incidence of the primary composite outcome was lower than anticipated (5.1% vs 7.0%), suggesting that the study may have

been underpowered for this outcome, as well as several of its important components (intrapartum stillbirth, neonatal mortality, Apgar score <4 at 5 minutes, neonatal encephalopathy, and neonatal seizures), which had incidences varying between 0.1% and 0.5%. Reasons for admission to the neonatal nursery may not necessarily reflect complications related to intrapartum hypoxia. Second, the lower incidence of the primary outcome in the sildenafil citrate group at the largest recruiting center (1.8%) compared with that at other sites (9.0%) that recruited substantially fewer women suggests that unmeasured heterogeneity in recruitment, conduct, or outcome measurement for the trial might be responsible for the absence of a treatment effect like the one observed in the earlier phase 2 trial.²⁰ Because this was a pragmatic trial, some variation in specific intrapartum management algorithms between study sites and between individual clinicians was expected and allowed within the protocol, which may have accounted for site differences particularly regarding decisions of when to intervene for suspected fetal compromise or management of postpartum hemorrhage. At the Mater Mother's Hospital, there is 24-hour on-site specialist obstetrician cover for the labor ward in contrast to other sites, where specialist obstetricians were normally off-site at night. Nevertheless, although there was a lower rate of the primary outcome at the Mater Mother's Hospital compared with other sites, there was no evidence of a differential treatment effect by site (test for interaction $P = .16$). There was no reduction observed in operative birth for fetal compromise even at the Mater Mother's hospital, and for the trial as a whole, there was no difference in fetal distress rates between both groups. Third, we do not have pharmacokinetic information, making it difficult to be certain whether the lack of effect on outcomes was due to inadequate maternal levels of sildenafil citrate or for other drug-related reasons. Fourth, because more than 80% of women in this trial underwent induction of labor, it is not possible to generalize these findings to all women attempting vaginal birth. Fifth, given the low incidence of adverse neonatal outcomes secondary to intrapartum hypoxic injury in high-income settings, reliably detecting a moderate but clinically important 10% reduction in poor neonatal outcome by any intervention, pharmacologic or otherwise, may require randomization of more than 75 000 participants.^{20,41}

Conclusions

Maternal oral sildenafil citrate during term labor did not result in a lower incidence of adverse perinatal outcomes potentially related to intrapartum hypoxia or in a reduction in emergency operative birth for intrapartum fetal compromise. These results, however, do not preclude its investigation in other areas of pregnancy care and different populations. Future research should focus on better identification of at-risk infants, alternative intrapartum therapies to mitigate complications related to hypoxia during labor, and consideration of mode and timing of birth to improve perinatal outcomes for late-gestation infants.

ARTICLE INFORMATION

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Author Affiliations: Mater Mothers' Hospital and Mater Research Institute—University of Queensland, Brisbane, Queensland, Australia (Kumar, Flenady, Liley); National Health and Medical Research Council Stillbirth Centre for Research Excellence, Brisbane, Queensland, Australia (Kumar, Flenady); Faculty of Medicine, University of Queensland, Brisbane, Queensland, Australia (Kumar); NHMRC Clinical Trials Centre, University of Sydney, Sydney, New South Wales, Australia (Tarnow-Mordi, Seidler, Simes, O'Connell); Department of Obstetrics and Gynecology, Monash University, Melbourne, Victoria, Australia (Mol); Cerebral Palsy Alliance, Sydney, New South Wales, Australia (Badawi); Department of Neonatology, University of Sydney, Sydney, New South Wales, Australia (Badawi); Obstetrics and Gynecology, Mercy Hospital, University of Melbourne, Melbourne, Victoria, Australia (Walker); Department of Obstetrics and Gynecology, Liverpool Hospital, Sydney, New South Wales, Australia (Hyett); Department of Obstetrics and Gynecology, Western Sydney University, New South Wales, Australia (Hyett); Department of Child and Adolescent Psychiatry, University Medical Center Rostock, Rostock, Germany (Seidler); German Center for Child and Adolescent Health (DZKJ), Greifswald/Rostock, Rostock, Germany (Seidler); School of Public Health, University of Technology Sydney, Sydney, New South Wales, Australia (Callander).

Author Contributions: Drs Kumar and O'Connell had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Kumar and Tarnow-Mordi contributed equally to the manuscript. Drs Simes and O'Connell served as co-last authors.

Study concept and design: Kumar, Tarnow-Mordi, Mol, Flenady, Liley, Badawi, Hyett, Seidler, Callander, Simes, O'Connell.

Acquisition, analysis, or interpretation of data: Kumar, Tarnow-Mordi, Liley, Badawi, Walker, Seidler, Callander, Simes, O'Connell.

Drafting of the manuscript: Kumar, Tarnow-Mordi, Flenady, Badawi, O'Connell.

Critical review of the manuscript for important intellectual content: Kumar, Tarnow-Mordi, Mol, Flenady, Liley, Walker, Hyett, Seidler, Callander, Simes, O'Connell.

Statistical analysis: Tarnow-Mordi, Mol, O'Connell.
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Review and interpretation of statistical analysis: Simes.

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Group Information: The iSEARCH investigators are listed in [Supplement 4](#).

Data Sharing Statement: See [Supplement 5](#).

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Additional Information: A trial management committee and an independent data and safety monitoring committee monitored the trial. The trial statistician had full access to all the data.

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